

Getting Started of Modular Application Creator

Version V18 12/2022

https://support.industry.siemens.com/cs/ww/en/view/109762852

Siemens Industry Online Support

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1 Introduction

The Modular Application Creator provides the automated generation of TIA Portal projects based on predefined and tested software modules. Especially in the case of complex machine configurations, such as multi belt control or other multi-axis applications, this is very efficient compared to manually creating and parameterizing the automation project directly in the TIA Portal.

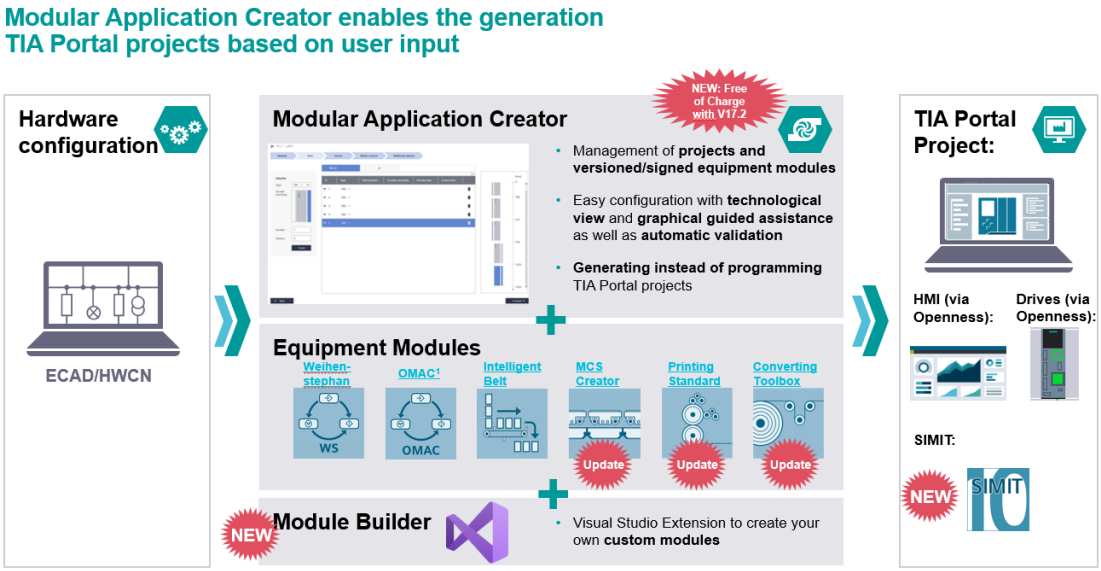
The Modular Application Creator (“MAC”) consist of two parts:

- The “MAC” software itself
- Equipment Modules to be used by the “MAC”

Both packages are handled independently but must be connected. As prerequisite to generate TIA Portal projects, **an installed TIA Portal V18 and an installed StartDrive V18 with activated Openness interface each** is needed. Details about this you will find in this SIOS article 109773802 in the chapter Installation: "Add user to the user group "Siemens TIA Openness"".

To use the generated software, a specific license is needed when used at a real machinery. The following picture shows the relationships:

Figure 1-1



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2 Download and first starting

To use the Modular Application Creator, download the Modular Application Creator all-in-one package from SIOS and unzip the downloaded file into another directory.

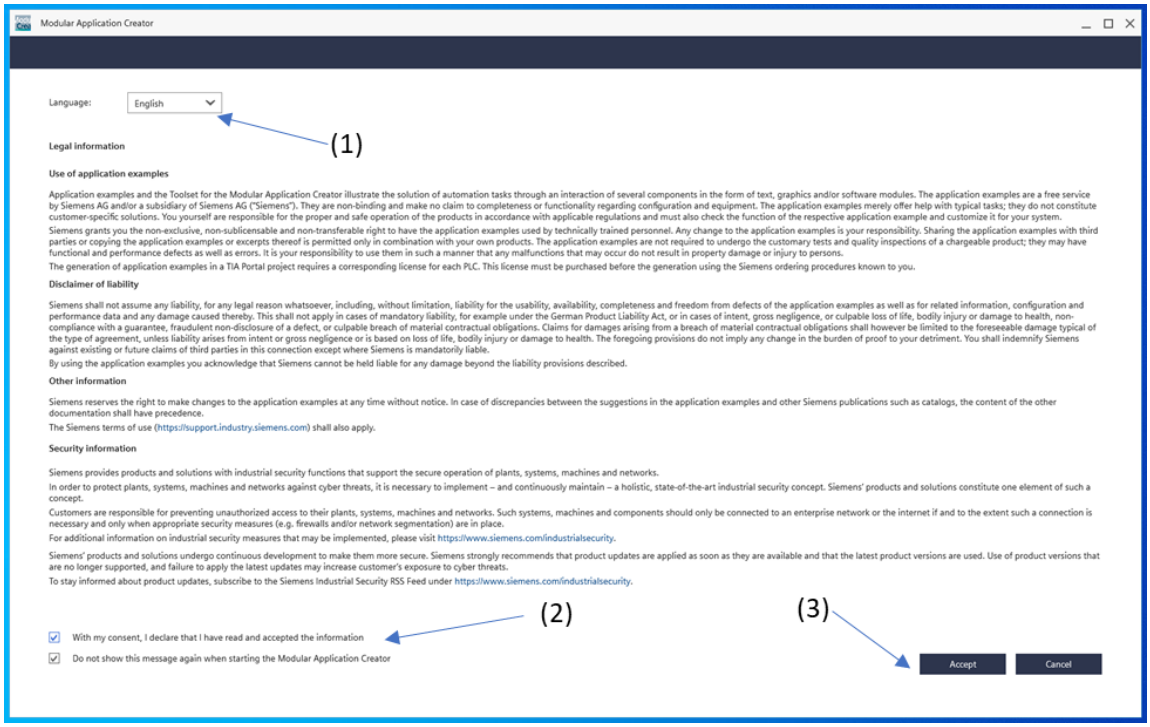
To run the software, execute the ModularApplicationCreator.exe, which is included in the package of the download of the Modular Application Creator.

Initially a disclaimer with legal information is shown. On the top the language can be changed of the text between german and english (1).

Please read carefully and confirm, if you are agree to all given terms and conditions, by activating the checkboxes (2) at the bottom.

Click “Accept” (3) to continue.

Figure 2-1



When accepted the disclaimer, the following window will appear to link the directory containing the equipment modules to the system component.
(with "all-in-one download" the path is preset)

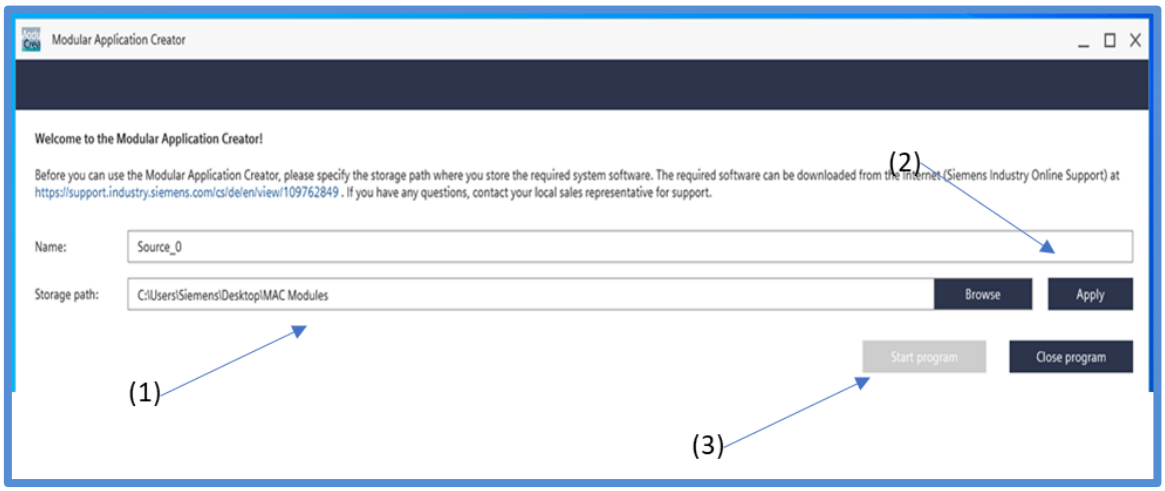
First enter the path of the directory, where the **system component** has been unzipped (1).

Then apply this path (2).

When this action has successfully identified all needed components, the program can be started (3).

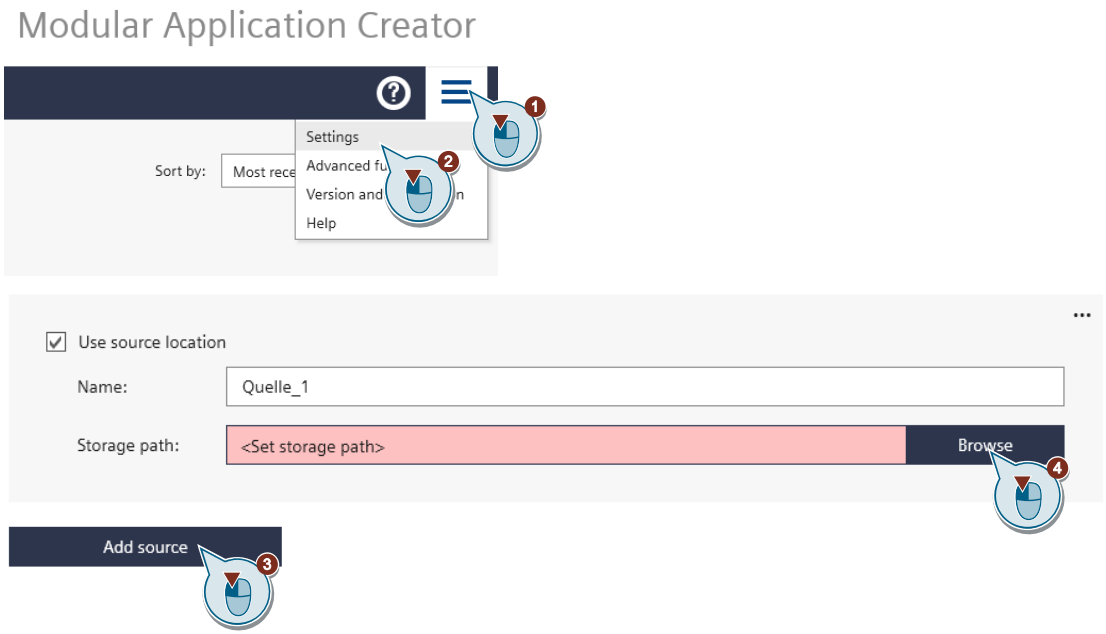
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Figure 2-2: selecting system variant



In order to use the Modular Application Creator, it must be linked to the equipment modules (not necessary if you use “all-in-one package) (see Figure 2-3). To do this, go to the settings (1) and (2). Click on "Add source" (3) and then select the location of the equipment modules (4). Select the folder with the equipment modules from the unzipped all-in-one download in the folder “packages”.

Figure 2-3 Connecting the Modular Application Creator and the Equipment Modules

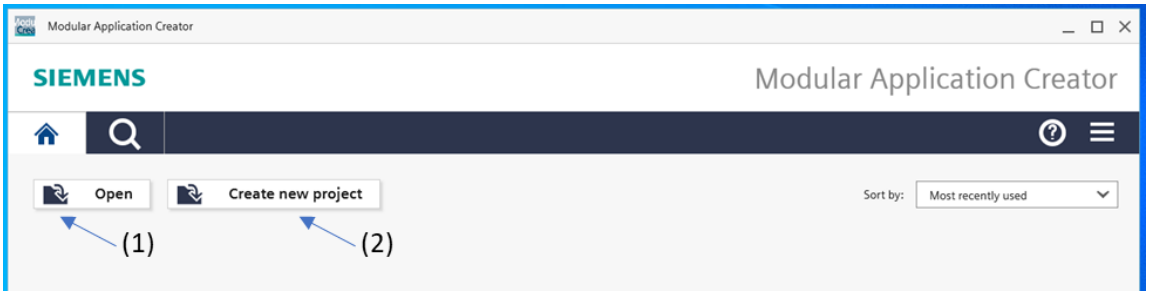


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3 Create the first project

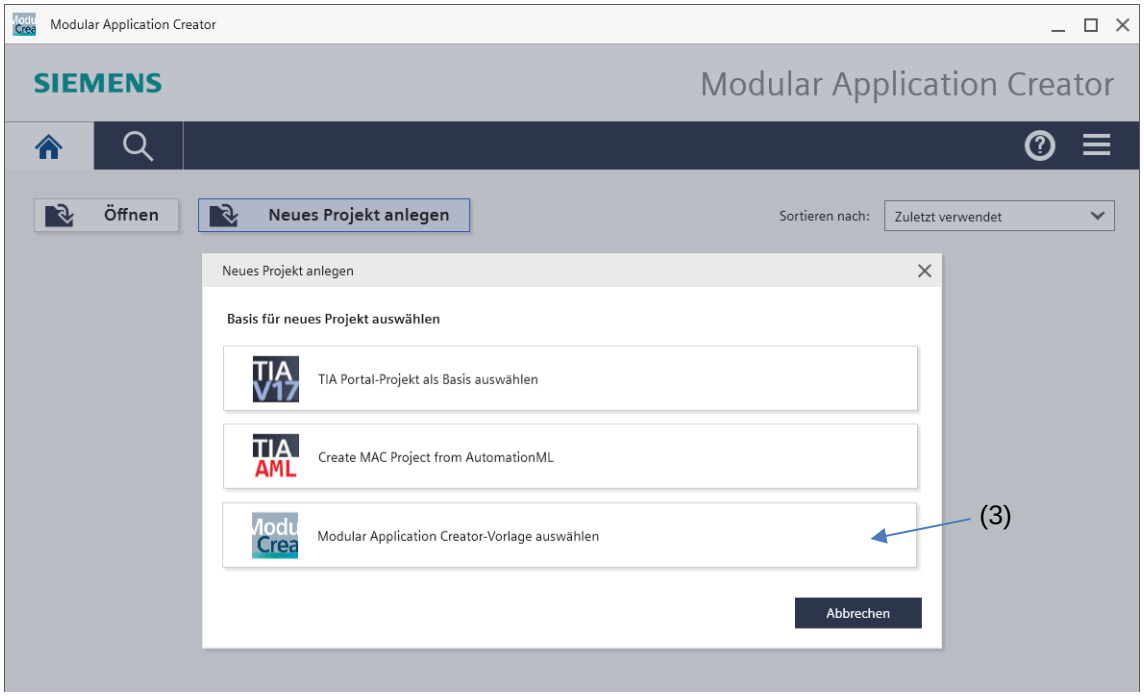
When the software starts, an empty screen appears with two options which allows to open an existing Modular Application Creator Project (1) or to create a new project (2).

Figure 3-1



To create the first “MAC” Project, the “Create new project” (2) should be selected. Then it can be started either with any TIA Portal project, based on version V18, or, just better for the first try, with a predefined project template included in the package of the modules. With this, select the second option (3).

Figure 3-2:



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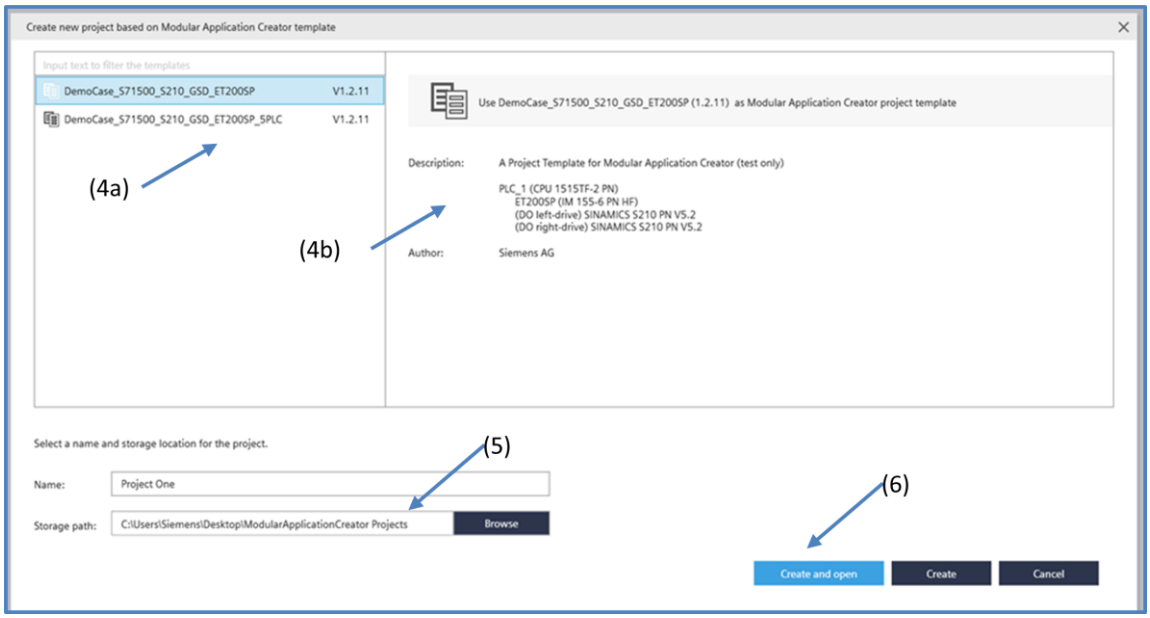
The window that then appears shows the list of the given project templates (4).

If one of the entries is selected, additional information on the template is displayed on the right (4b)

Before creating a new “MAC” project, enter a meaningful project name and select a preferred storage path (5).

Select "Create and Open" (6) to close the window and open the new project directly

Figure 3-3



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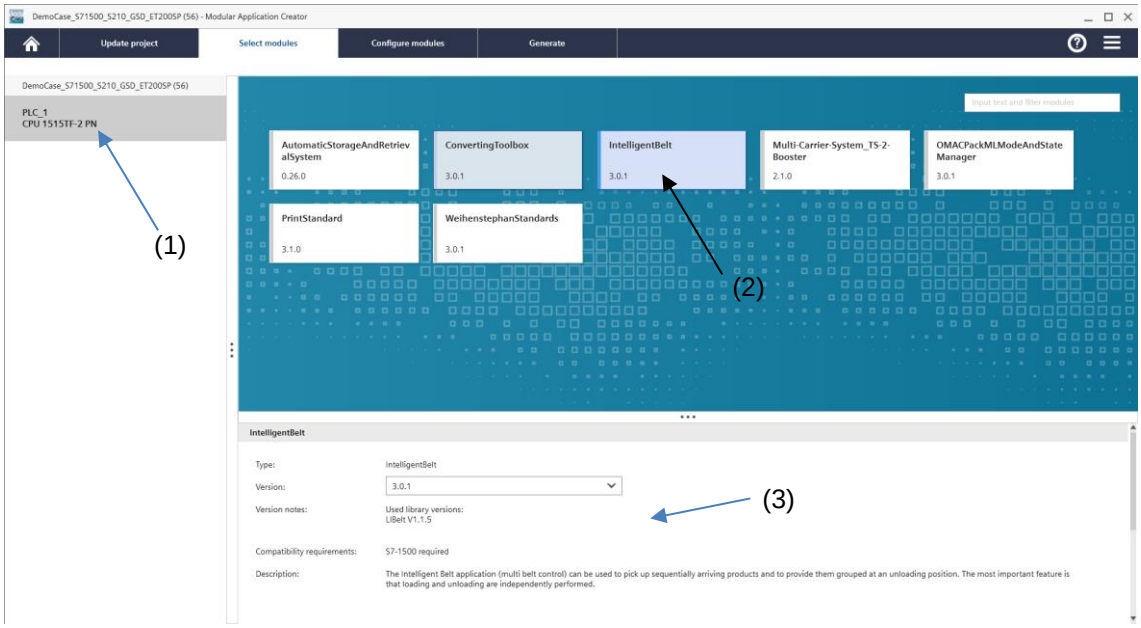
4 Select an equipment module and add it to the project

In the next step the modules have to be selected and assigned to a PLC.

If the “Create and open” was selected, the project opens directly and shows the next screen. On the left side the available PLC in the used TIA Portal project is shown (1) and in the middle all equipment modules, which the software has identified in the initially defined folder.

By selecting a module (2), additional information concerning to this module is shown in the lower part of the screen (3). Choose e.g. the Intelligent Belt module to use them in the further workflow. The Intelligent Belt module represents a multi belt control.

Figure 4-1



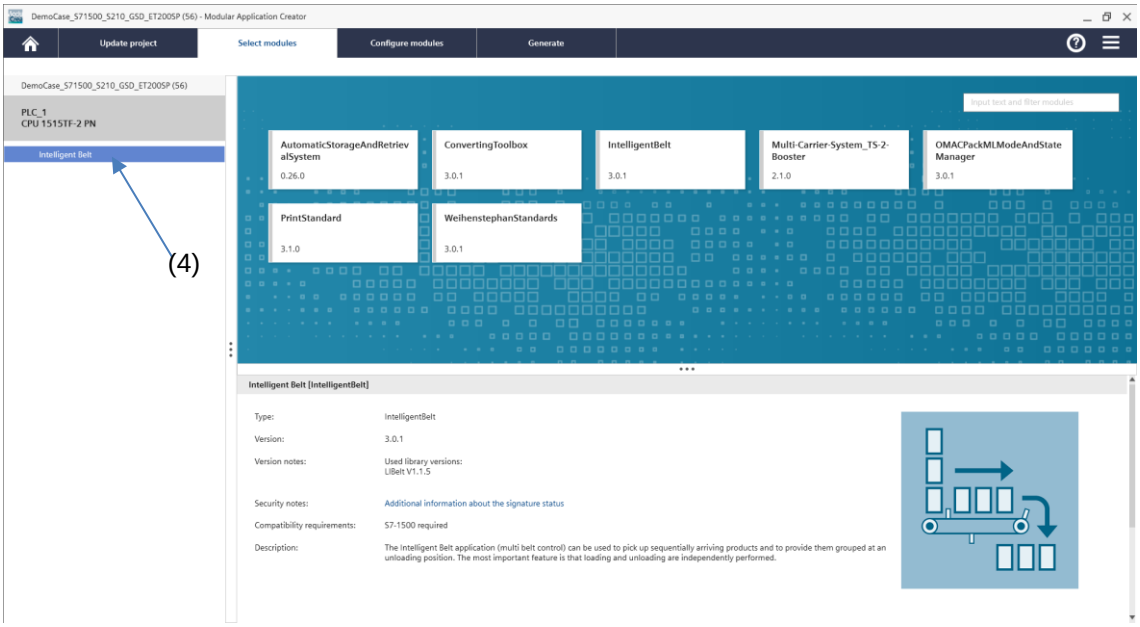
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To use a module, it must be assigned to that PLC into which it will be generated in the TIA Portal project. The assignment is done by dragging and dropping or by using the right mouse button.

While a module is being assigned, the system checks whether the module is valid for use with this PLC (for example, if the module requires a Technology PLC). This early validation helps to detect compatibility problems before a non-working generation would be executed.

If the module is assigned to a PLC, it is requested to give this specific instance of the module a meaningful name (4). This name will be used for the generation and affects the generated code and resources also.

Figure 4-2



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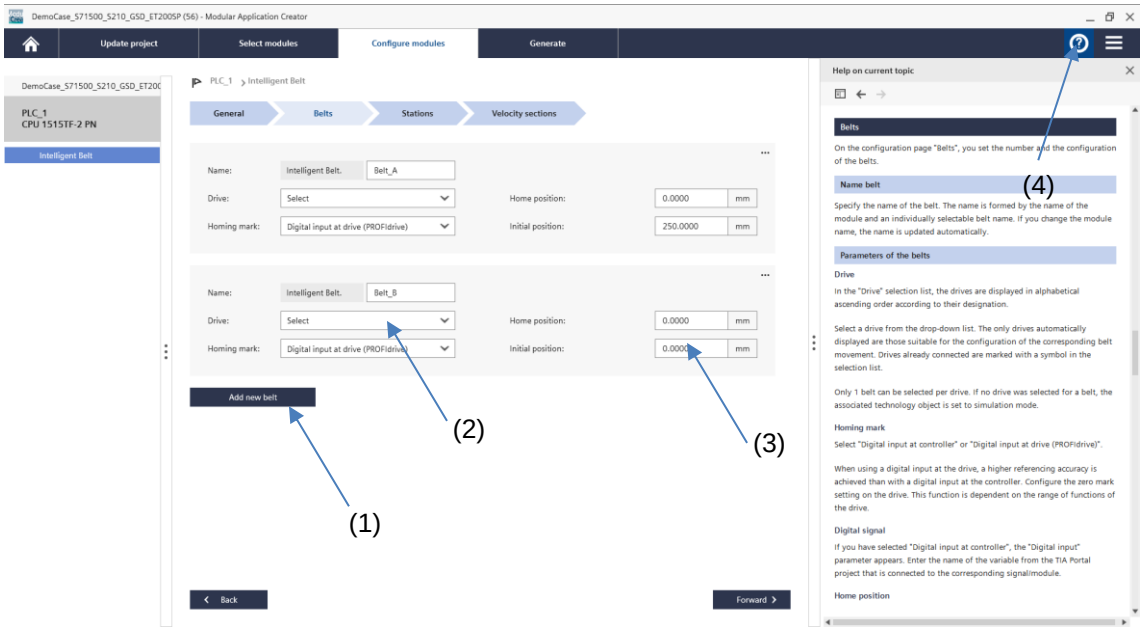
5 Configure an equipment module

When the module is assigned to a PLC it can be configured in detail.

To do this, select the module and open the configuration mode with either double clicking or choose the button “Configure Modules” in the headline of the software.

The screens then presented are individual for each equipment module and show the underlying knowledge from a technological perspective. The configuration may contain several pages and you can be navigate using the internal tabs at the top of the configuration or using the "Forward" and "Back" buttons at the bottom.

Figure 5-1



To generate a valid TIA Portal project, first the two axes (1) have to be named, then the corresponding drives have to be selected (2) and finally some values can be set (3). These values have an internal validation to limits or wrong entries. Just give it a try!

Equipment modules can provide an online help concerning the contained technology. The help window can be activated with the “?” button (4) or by pressing the F1 key. Besides this, the help for the tool itself is also available with these actions.

Please note: All configurations you made are stored automatically into the project files. Switching between screens or modules or close and reopen the project later on is possible without loss of data.

Since all data are configured the last step is ready to start, generating the resulting TIA Portal project.

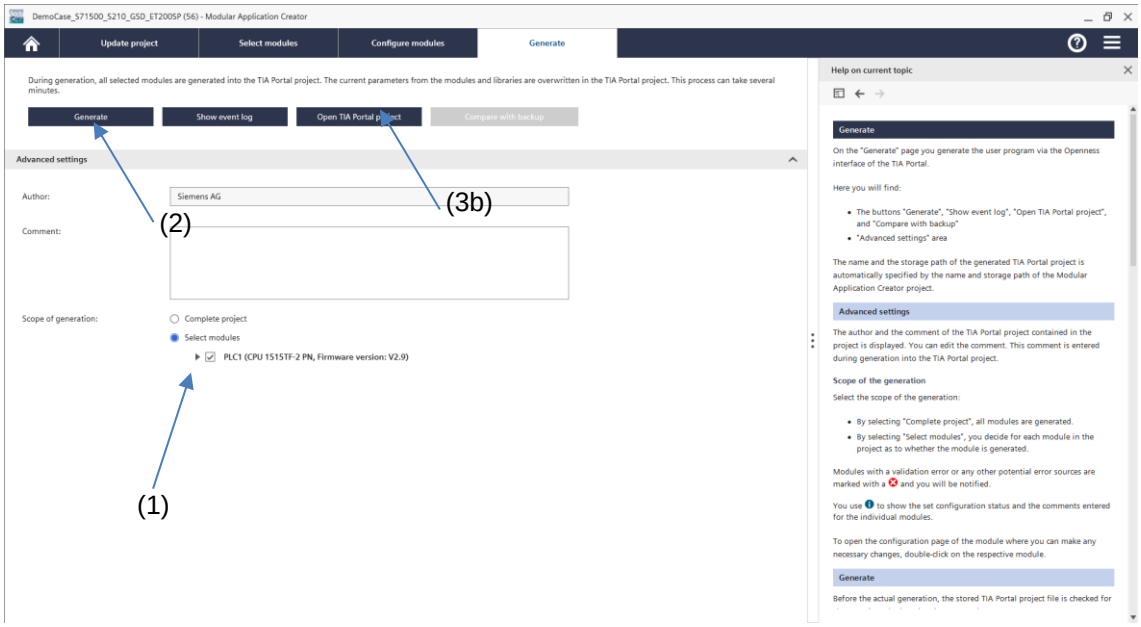
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6 Generate the corresponding TIA Portal project

To do this the generation page has to be opened by selecting the “Generate” tab in the navigation headline of the software.

To execute a generation, an installed TIA Portal V18 and an installed StartDrive V18 with each activated Openness Interface is needed.

Figure 6-1



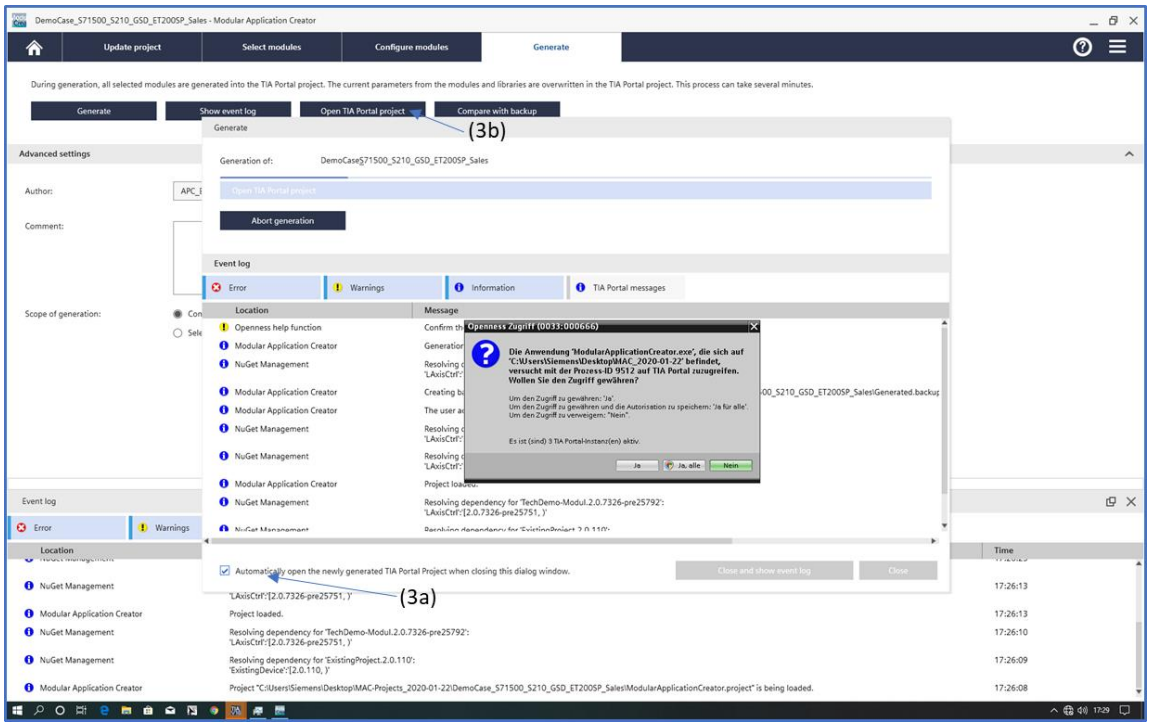
In the “Advanced Settings” a comment for the TIA Portal project can be added and the scope of generation can be modified also (1). E.g. specific modules or PLCs can be excluded from the generation process.

Push the button “Generate” (2) to start the generation process into TIA Portal. Please be aware that the generation will need some minutes, but you can monitor the action in the then upcoming EventLog.

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An Event Log window will appear to show and log different status information during the generation process. Please note that the software is then in a modal mode, means that it cannot be operated until the generation process is aborted, or the generation is finished.

Figure 6-2



Note Please be aware, that if not done before, the Openness firewall dialog of the TIA Portal Openness interface will come up automatically and has to be accepted explicitly to start the generation process.

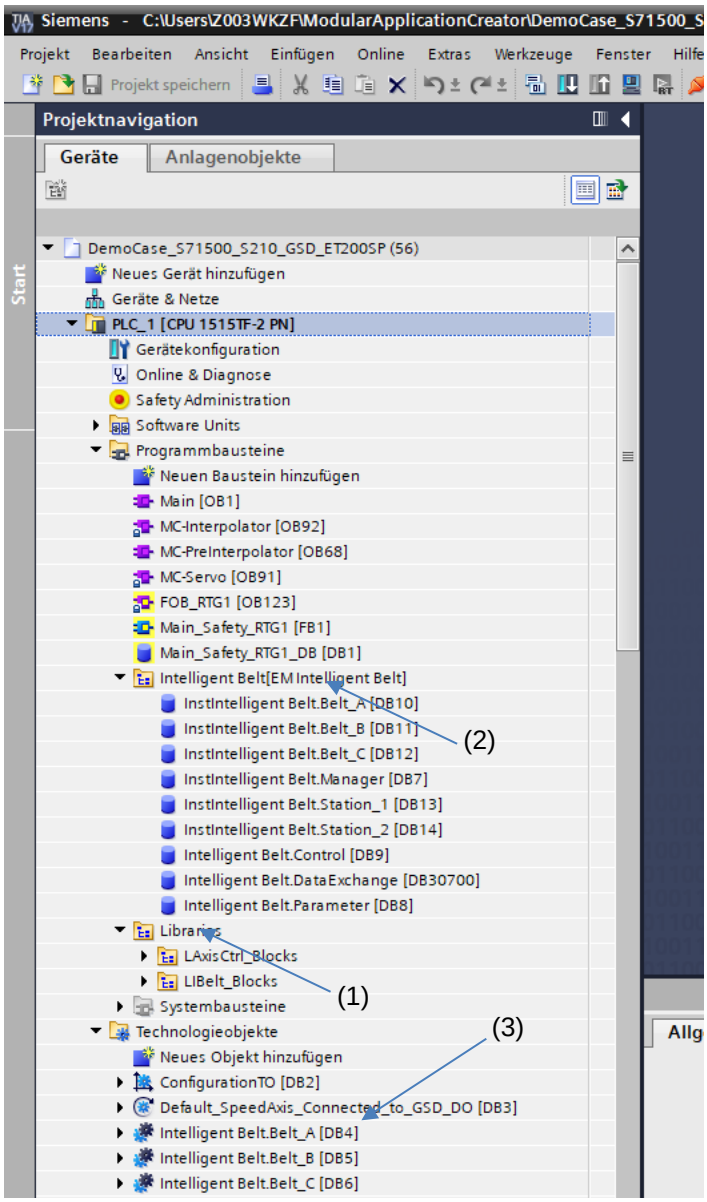
When the generation is finished the TIA Portal project can be opened directly with accepting the checkbox in the left corner of the Event Log window (3b) or afterwards with pushing the button “Open TIA Portal Project” (3a) on this page.

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7 Check the generated TIA Portal project

When selecting “Open TIA Portal Project” the TIA Portal project will be opened and all the generated objects can be found in there, e.g. libraries (1), the generated group of the module, including all the instance data blocks (2) and even the generated technology objects (3) including all the parameters which have been configured just before in the Modular Application Creator:

Figure 7-1



With this, the project is ready to compile and to download to the real PLC or to the PLCSIM and to test it.

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8 Appendix

8.1 Service and support

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8.3 Application support

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8.4 Links and literature

Table 8-1

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109762852

8.5 Change documentation

Table 8-2

Version	Date	Modifications
V2.0	02/2020	First version
V2.1	04/2021	Refinement in new format
V3.0	09/2021	Adjusted to MAC 3.0
V17.1	04/2022	Adjusted to MAC 17.1
V17.2	07/2022	Adjusted to MAC 17.2

Version	Date	Modifications
V18	12/2022	Adjusted to MAC 18

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